Freestanding Magneto-Mechanical Metamaterials

Ariana Olivares, Jay Sim, Sarah Hwang and Renee Zhao

Department of Mechanical Engineering, Stanford University

SIM LAB

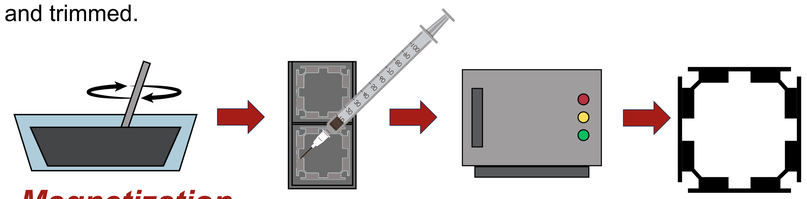
Background and Motivation

Metamaterials are special materials that can change shape quickly and in a controlled way. Magneto-mechanical metamaterials do this by responding to magnetic fields. Hard-magnetic soft materials are made by mixing magnetic particles into a soft polymer, allowing them to bend and move without wires or motors. Past work demonstrates the flexibility of 2D magneto-metamaterials in achieving controllable performance.[1, 2].

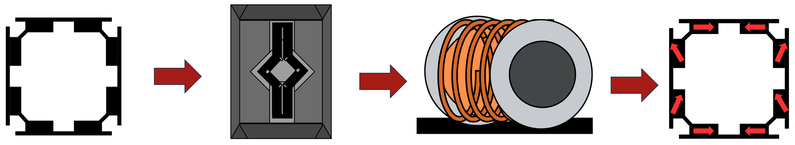
In this work, we study freestanding metamaterials using out-of-plane magnetization. We use experiments and simulations to understand how they respond to magnetic fields and explore their potential in soft robotics, reconfigurable devices, and adaptive structures.

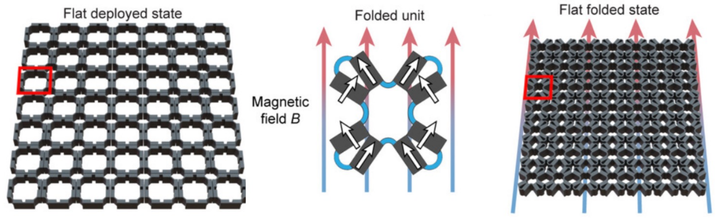
Fabrication

To fabricate the metamaterials, we adapted a design and fabrication process from previous work [1, 2]. A Dragon Skin 20 silicone elastomer blend mixed with NdFeB magnetic particles was degassed and poured into 3D-printed PVA molds, heated at 80 °C for 30 minutes. After curing, unit cells were extracted and trimmed.

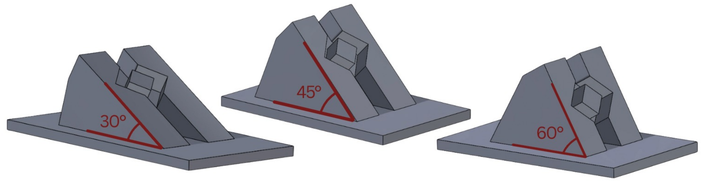


Magnetization



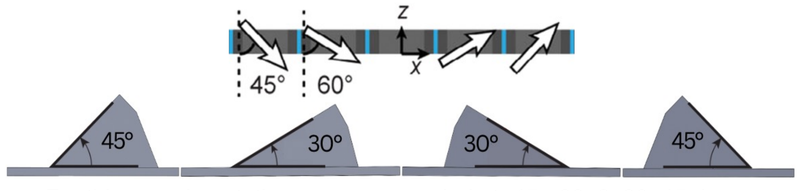


Custom fixtures oriented unit cells for magnetization at any target angle. Arranged in a diamond pattern, they were magnetized in a 1.5 T impulse field. By adjusting fixture orientation, we precisely control the magnetization direction, enabling out-of-plane configurations.



Assembly

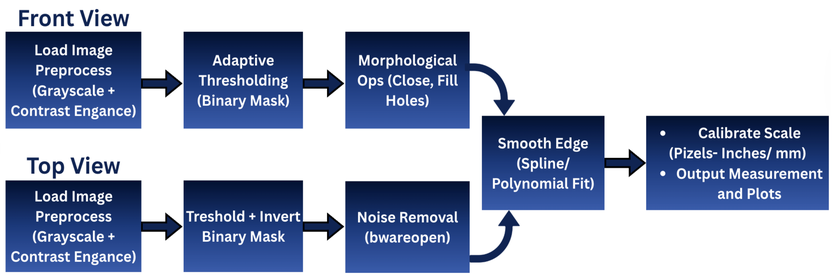
Fixture Order



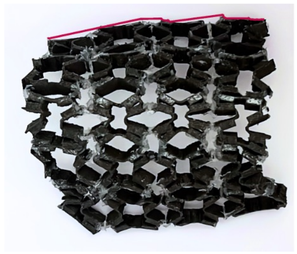
For this experiment, the rows were angled at -45, -30, 0, 30, 45 degrees

Experimental

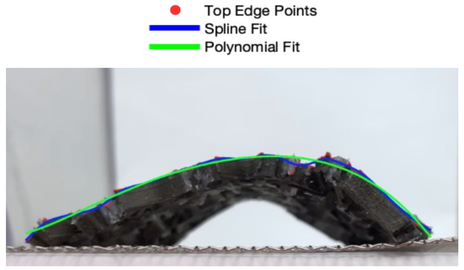
MATLAB Flow Chart



Top View Edge Fit



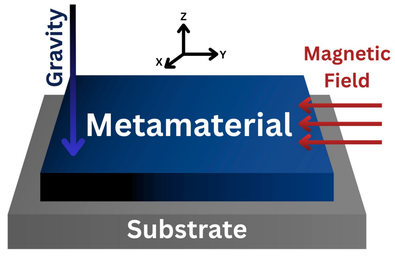
Front View Edge Fit



Simulation

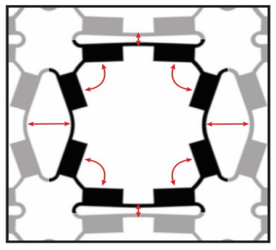
Abaqus 2021 was used to simulate the magnetic transformation of the metamaterial.

Boundary Conditions (BCs)



- **Magnetic Field** – Applied at an angle to each unit cell.
- **Gravity** – Downward load matching experimental conditions.
- **Fixture/Substrate** – Substrate is fixed; unit cells are free except at contact.

Contact

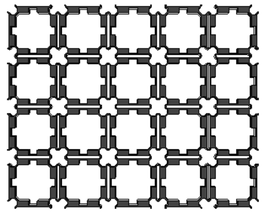


- Surface-to-surface interaction with friction to capture sliding resistance.
- Contact areas shown in black; red arrows indicate interaction forces.

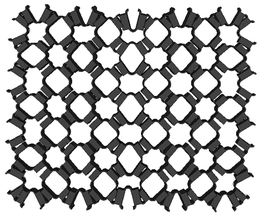
Front View



Flat Deployed State



Flat Folded State

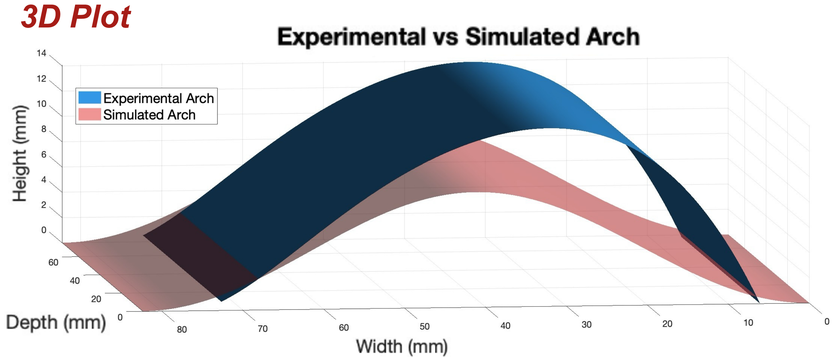


Results

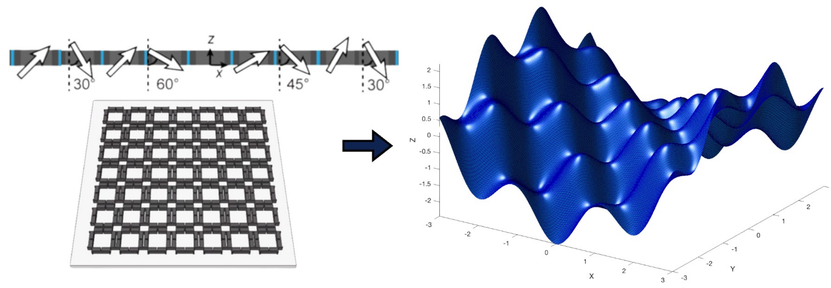
The simulation closely replicated the experimental folding patterns, vertical rise, and bending profiles, though the simulation showed slightly lower upward displacement, resulting in a somewhat greater width compared to the experiment. These differences likely stem from variations in friction between the metamaterial and substrate, as well as manufacturing tolerances. Overall, the simulation captures the dominant mechanics of the system.

	Experimental	Simulation
Width	70 mm	82.31 mm
Height	13.17 mm	9.10 mm
Depth	76.28 mm	75.25 mm

3D Plot



This project is conducted in collaboration with Prof. Wei Chen at Northwestern University, aiming to create design-optimized metamaterials capable of morphing from flat sheets into complex 3D geometries. Planned developments include algorithms that assign precise magnetization angles to each unit cell, enabling controlled shape formation. To improve the accuracy of future models, we will integrate refined friction behavior, higher-resolution contact meshing, and manufacturing tolerance effects into the simulation framework.



Results and Conclusion

This research demonstrates control of freestanding metamaterials with out-of-plane magnetization through experiments and simulations. In collaboration with Northwestern University, we validate and improve models of complex magnetized metamaterials. Our results support future applications in soft robotics, reconfigurable devices, and adaptive structures.

Acknowledgments

I would like to thank Professor Zhao and the Zhao Lab for welcoming me over the summer, Jay Sim for his mentorship and guidance, and Stanford SURF for providing this research opportunity.

References

[1] Wu, X., et al. "Magnetically Actuated Reconfigurable Metamaterials as Conformal Electromagnetic Filters." *Advanced Intelligent Systems*, vol. 4, no. 12, 2022, 2200106. https://doi.org/10.1002/aisy.202200106

[2] Kim, Y., et al. "Magneto-Mechanical Metamaterials with Widely Tunable Mechanical Properties and Acoustic Bandgaps." *Advanced Functional Materials*, vol. 33, no. 2, 2023, 2303204. https://doi.org/10.1016/j.actamat.2023.119279